Subject: Biology and Environmental Science (CY202)

Question Bank

Unit 3: Structure and Function of Ecosystem

SECTION A — Two Mark Questions

ENVIRONMENT AND ECOSYSTEM

Q1. What is the difference between environment and ecosystem? [2]

Answer: The environment is the surrounding in which living organisms live — everything around us, including living organisms and non-living things such as soil, air and water. The ecosystem is an organized system in which living organisms interact with the environment; it is the organized community of living beings where biotic and abiotic elements are in continuous interaction with each other.

ECOSYSTEM

Q2. Define ecosystem. [2]

Answer: Ecosystem is the major ecological unit; it is the basic functional unit of organisms and their environment interacting with each other. It has both structure (related to species diversity) and function (related to energy flow, decomposition, nutrient cycling and major biomes).

STRUCTURE OF ECOSYSTEM

Q3. Mention the types of components of the ecosystem with examples. [2]

Answer:

- **Abiotic component:** Includes basic inorganic substances (soil, water, oxygen, calcium carbonates, phosphates) and organic compounds, plus physical factors such as moisture, wind currents and solar radiation. Radiant energy of the sun is the only significant energy source for any ecosystem.
- **Biotic component:** Includes producers (autotrophs, e.g. green plants), consumers (herbivores/carnivores, e.g. rabbit, deer, cow, cats, dogs, lion, tiger) and decomposers (bacteria and fungi that degrade dead remains).

LEVELS OF ORGANIZATION

Q4. What are the levels of organization in the ecosystem? Mention the hierarchy. [2]

Answer: The hierarchy of organization is: Individual → Population → Community → Ecosystem → Biome → Biosphere. Each level represents an increasing scale, from a single organism to the entire part of the Earth in which life exists.

Q5. Arrange the following in correct order of organization: population, community, individual, ecosystem, biosphere, biome. [2]

Answer: Individual → Population → Community → Ecosystem → Biome → Biosphere.

ENVIRONMENTAL SUSTAINABILITY

Q6. What do you mean by environmental sustainability? [2]

Answer: Environmental sustainability is the responsibility to conserve natural resources and protect global ecosystems to support health and wellbeing, now and in the future. Because many decisions that impact the environment are not felt immediately, a key element of environmental sustainability is its forward-looking nature (e.g., India's Namami Gange Programme for conservation and rejuvenation of the River Ganga).

PRIMARY PRODUCTION

Q7. What are NPP and GPP? What is their relation? [2]

Answer:

- **GPP:** (Gross Primary Production) is the total amount of chemical energy (or biomass) stored by plants per unit area per unit time.

- **NPP:** (Net Primary Production) is the remaining production after respiration, used in plant growth and reproduction.

Net Primary Production (NPP) = Gross Primary Production (GPP) − Respiration

NPP is typically 80–90 per cent of GPP levels.

ENERGY FLOW

Q8. State the laws of thermodynamics followed in energy flow in ecosystem. [2]

- **First Law of Thermodynamics:** Energy can neither be created nor destroyed; solar energy can only be transferred from one system to another or converted from one form to another.
- **Second Law of Thermodynamics:** During every energy transformation, a portion of the energy is dissipated into the surroundings as heat energy — hence energy flow through an ecosystem is unidirectional (not cyclic).

NUTRIENT RECYCLING

Q9. What do you mean by nutrient recycling? [2]

Answer: Nutrients are substances required by organisms for growth, development, reproduction and survival. They are continuously used by organisms and returned to the environment in a modified form, which is converted back to its original form by other organisms. This cyclic flow of nutrients from the environment to organisms and back into the environment is called nutrient recycling.

BIOGEOCHEMICAL CYCLE

Q10. Which biogeochemical cycle does not directly involve air or atmosphere? [2]

Answer: The Phosphorus cycle. Unlike the carbon, nitrogen, sulphur, oxygen and hydrologic cycles (which have a significant gaseous/atmospheric phase), phosphorus has no notable gaseous phase — it is a sedimentary cycle in which phosphorus moves mainly through rocks, soil and water via weathering, erosion and sedimentation.

FOOD CHAIN

Q11. What is food chain? Explain in brief with schematic diagram. [2]

Answer: A food chain is the straight chain of organisms that starts with the producer, through which energy is transferred to the carnivores, and ends with the decomposer in the food web. It explains the feeding relationship of organisms — producers (autotrophs) → primary consumers (herbivores) → secondary/tertiary consumers (carnivores) → decomposers.



Fig: A simple food chain — producer to final consumer/decomposer.

NITROGEN CYCLE

Q12. What do you mean by nitrogen fixation? [2]

Answer: Nitrogen fixation is the conversion of inert atmospheric nitrogen gas (N2) into usable chemical forms such as ammonia/ammonium. It is carried out mainly by nitrogen-fixing bacteria (e.g., Rhizobium in root nodules,

and free-living bacteria in soil), and also occurs naturally through lightning and volcanic activity, or industrially through fertilizer manufacture.

Q13. What do you mean by nitrification in nitrogen cycle? [2]

Answer: Nitrification is the microbial oxidation of ammonium (NH_4^+) into nitrite (NO_2^-) and then into nitrate (NO_3^-) by nitrifying bacteria present in soil and water — making nitrogen available in a form usable by plants.

Q14. What do you mean by assimilation in nitrogen cycle? [2]

Answer: Assimilation is the process by which plants take up inorganic nitrogen (ammonium and nitrate) from the soil through their roots and incorporate it into organic molecules such as proteins and nucleic acids. Animals obtain their nitrogen by consuming these plants (or other animals).

Q15. What do you mean by ammonification in nitrogen cycle? [2]

Answer: Ammonification is the process by which decomposers (bacteria and fungi) break down nitrogenous waste (such as urine) and dead plants/animals, releasing ammonium (NH_4^+) and ammonia (NH_3) back into the soil.

Q16. Write down four importance of nitrogen cycle. [2]

- Supplies usable nitrogen required for the synthesis of proteins and nucleic acids in living organisms.
- Maintains soil fertility by continuously replenishing usable nitrogen compounds.
- Regulates key ecosystem processes such as primary production and decomposition.
- Human activities (fossil fuel combustion, artificial fertilizers, wastewater) have dramatically altered the cycle, which can negatively affect the environment and human health if not properly managed.

PHOSPHORUS CYCLE

Q17. What do you mean by weathering in Phosphorus cycle? [2]

Answer: Weathering refers to the gradual breakdown and erosion of phosphate-containing rocks by physical and chemical processes, which releases phosphate into the soil and, via runoff in rivers and streams, into water bodies. Since phosphorus has no significant atmospheric/gaseous phase, weathering of rocks is the primary natural source of phosphate for uptake by plants and aquatic organisms.

ENVIRONMENTAL GRADIENT

Q18. What do you mean by environmental gradient? [2]

Answer: An environmental gradient is the gradual change in the level of a physical or chemical environmental factor (such as light, temperature, pH or altitude) between two extremes — high to low, or large to small (e.g., a global temperature gradient from the equator towards the poles). Each species has a tolerance range within which it can function efficiently along a given gradient.

SECTION B — Seven Mark Questions

BIOTIC COMPONENTS

Q1. Discuss the biotic components of the ecosystem and their organization. [7]

Answer: Biotic components are the living parts of an ecosystem. They include producers, consumers and decomposers:

- **Producers:** Autotrophic, chlorophyll-bearing organisms such as plants, which produce their own food through photosynthesis.
- **Primary Consumers:** Herbivores that get nutrition by eating plants directly, e.g. rabbit, deer and cow.
- **Secondary Consumers:** Carnivores that eat the flesh of herbivores, e.g. cats and dogs.
- **Tertiary Consumers:** Carnivores that prey upon other carnivores, e.g. lion, tiger and vulture.
- **Decomposers:** Attack the dead remains of producers and consumers and degrade complex organic substances into simpler compounds, playing a vital role in maintaining the dynamic nature of the ecosystem.

Organization within the biotic component follows a defined hierarchy:

- **Individual:** All individuals of a species potentially able to interbreed and produce fertile offspring (e.g., the salmon Salmo salar).
- **Population:** A group of individuals of the same species living in a particular area at the same time; each population is genetically distinct to some degree and has a size, birth rate, death rate and growth rate.

- **Community:** Populations of different species living together and interacting, forming food webs; communities occur in habitats determined by topography, vegetation structure, geology and the surrounding medium.
- **Ecosystem:** Encompasses both the living (biotic) and non-living (abiotic) components of an area, with strong interaction between them; major processes are nutrient cycling and energy flow.
- **Biome:** A group of organisms sharing common characteristics correlated with the environment they exist in; may span more than one continent (e.g., tropical rainforest, coniferous forest/taiga, grassland, desert scrub).
- **Biosphere:** The highest organizational level — that part of the Earth and atmosphere in which life exists, including the surface layer of land, oceans, sediments and the part of the atmosphere occupied by life.

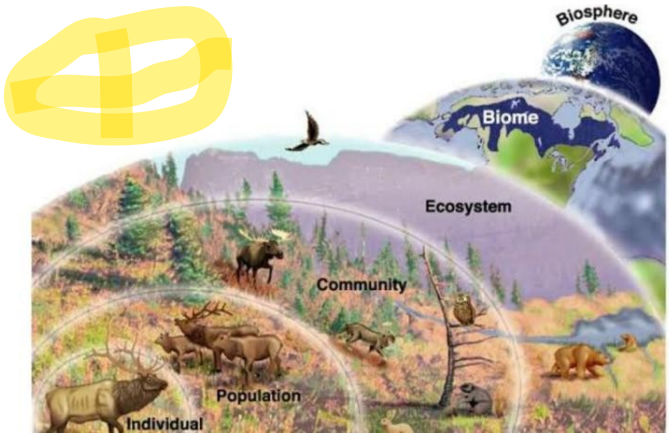


Fig: Levels of biological organization — individual, population, community, ecosystem, biome and biosphere.

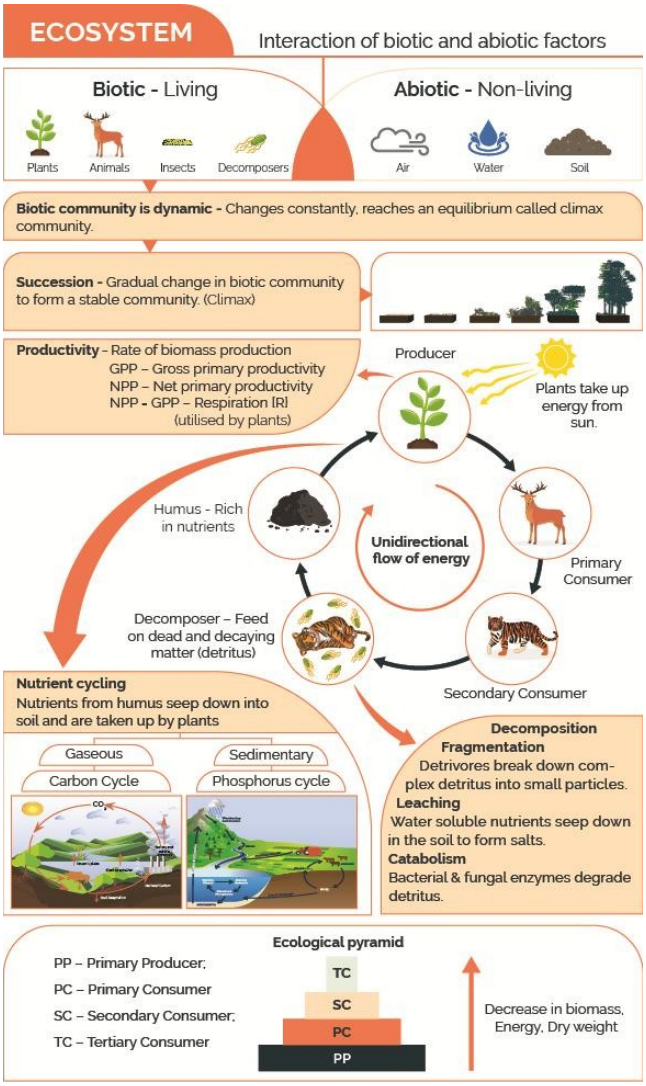


Fig: Interaction of biotic and abiotic factors in an ecosystem, showing productivity, nutrient cycling and the ecological pyramid.

FUNCTIONS OF ECOSYSTEM

Q2. Discuss in details about the functions of the ecosystem. [7]

Answer: The function of an ecosystem is related to energy flow, decomposition, nutrient cycling and major biomes.

- **Energy Flow:** A proportion of light energy is converted to chemical energy. There are two sources of energy — autotrophic (energy-rich organic matter produced within the ecosystem by green plants via photosynthesis) and heterotrophic (chemical energy imported as organic matter originating from primary production elsewhere, e.g. dead leaves entering a shaded forest stream).

$$12\ H2O + 6\ CO2 + 709\ kCal \rightarrow C6H12O6 + 6\ O2 + 6\ H2O \quad (Photosynthesis)$$

- **Primary Production:** Only 1–5% of total solar radiation is used in photosynthesis. Glucose produced is stored as starch or combined into carbohydrates like cellulose; with other inorganic substances (N, P, Mg, Fe) plants produce fats, proteins and nucleic acids. This production of organic matter by plants is called primary production (Gross Primary Production and Net Primary Production, where NPP = GPP − Respiration).
- **Nutrient Cycling:** The cyclic flow of nutrients from the environment to organisms and back — nutrients are continuously used and returned in modified form by decomposers.
- **Decomposition:** Detritivores break down complex detritus into small particles (fragmentation); water-soluble nutrients seep down into the soil to form salts (leaching); bacterial and fungal enzymes degrade detritus (catabolism) — releasing nutrients back for uptake by plants.
- **Succession:** A gradual change in the biotic community leading to a stable, self-perpetuating community called the climax community.
- **Major Biomes:** Large-scale ecological units defined by climate and dominant vegetation (e.g., tropical rainforest, taiga, grassland, desert scrub) — the biotic community is dynamic and constantly changes, eventually reaching this equilibrium.

TROPHIC LEVELS

Q3. What are trophic levels? Discuss each trophic level with example. [7]

Answer: A trophic level refers to the position an organism occupies in a food chain, based on its feeding relationships. Energy and biomass decrease progressively from one trophic level to the next.

- **First Trophic Level — Primary Producers:** Autotrophs / green plants that convert solar energy into chemical energy through photosynthesis, forming the base of the food chain. Example: grass, plants.
- **Second Trophic Level — Primary Consumers:** Herbivores that feed directly on producers. Example: grasshopper, ground squirrel, rabbit, deer, cow.
- **Third Trophic Level — Secondary Consumers:** Primary carnivores that feed on herbivores. Example: frog, scorpion, cats and dogs.
- **Fourth Trophic Level — Tertiary Consumers:** Secondary carnivores that prey upon other carnivores; these are typically the top of the food chain. Example: snake, kit fox, golden eagle, hawk, lion, tiger.
- **Decomposers:** Fungi and bacteria that break down dead organisms and waste from all trophic levels, returning nutrients to the soil. Example: fungi, earthworm.

At each successive trophic level, only about 10% of the energy absorbed by one level is passed on to the next (the rest is lost as heat via respiration), which is why food chains are typically limited to 4–5 levels and why the biomass/energy pyramid narrows towards the top.

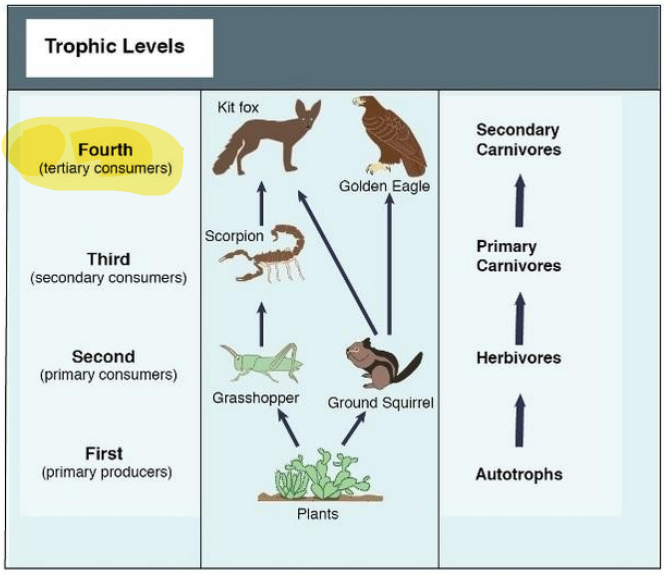


Fig: Trophic levels and food web — from primary producers to tertiary consumers.

ENERGY FLOW

Q4. How does energy flow in ecosystem? Discuss elaborately. [7]

Answer: Ecological energetics is the study of the movement of energy through an ecosystem.

- The sun provides all the energy that living beings require. Green plants absorb around 1% of total radiant energy for distribution across the ecosystem — despite its small size, this is adequate to maintain all life on Earth.
- Fuel enters the food chain at the level of producers in the form of solar energy. Plants convert solar energy into chemical energy through photosynthesis. This chemical energy is then passed from one trophic level to the next along a food chain.
- First Law of Thermodynamics: energy can neither be created nor destroyed — solar energy is only transferred from one system/form to another.
- Second Law of Thermodynamics: during every transformation, a portion of energy is dissipated into the surroundings as heat — hence energy flow is unidirectional, unlike nutrient cycling.

As shown in the energy pyramid, roughly 10% of the energy at one trophic level is transferred to the next (Primary Producers ≈10,000 Kcal/m²/year → Primary Consumers ≈1,000 → Secondary Consumers ≈100 → Tertiary Consumers ≈10 Kcal/m²/year), with the majority lost as heat through cellular respiration at every step. Decomposers live off detritus, decomposing waste and dead matter from every trophic level.

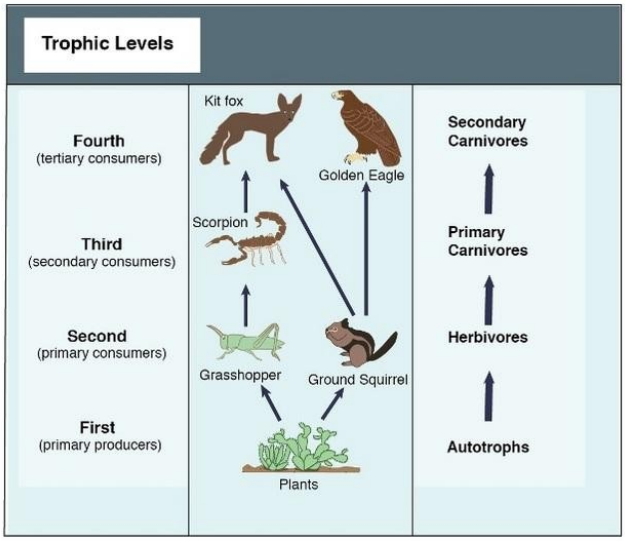


Fig: Energy transfer and loss across trophic levels (energy pyramid).

CARBON CYCLE

Q5. Discuss the carbon cycle in details. [7]

Answer: The carbon cycle is the part of the biogeochemical cycle by which carbon is exchanged among the biosphere, pedosphere, geosphere, hydrosphere and atmosphere of Earth.

- Carbon is the main component of biological compounds as well as a major component of many minerals. The cycle comprises a sequence of events key to making Earth capable of sustaining life, describing the movement of carbon as it is recycled and reused, as well as long-term carbon sequestration and release from carbon sinks.
- **Key processes:** CO_2 is fixed by plants and phytoplankton through photosynthesis (assimilation) and returned to the atmosphere through plant and animal respiration, combustion (of fossil fuels and biomass) and volcanic activity.
- Dead animals, plants and waste undergo organic decomposition; over geological time, this can lead to the formation of fossil fuels (peat, coal, oil).
- CO_2 dissolves directly from the atmosphere into oceans/lakes and in precipitation, forming carbonic acid (contributing to ocean acidity) — it may also be absorbed by rocks through weathering, or washed into the ocean as marine deposits of $CaCO_3$ (via assimilation by phytoplankton and marine sedimentation).
- Carbon exists mainly as CO_2 and methane (CH_4) in the atmosphere — both absorb and retain heat, contributing to the greenhouse effect. Methane has a larger greenhouse effect per volume than CO_2 but exists in lower concentrations and is shorter-lived, so CO_2 contributes more to the overall global greenhouse effect.
- **Human impact:** Human activities (land-use change, mining/combustion of fossil fuels, cement manufacture) have disturbed the fast carbon cycle for centuries. Atmospheric CO_2 had increased nearly 52% over pre-industrial levels by 2020, forcing greater atmospheric/surface heating and reducing ocean pH — fundamentally altering marine chemistry and contributing to human-caused climate change.

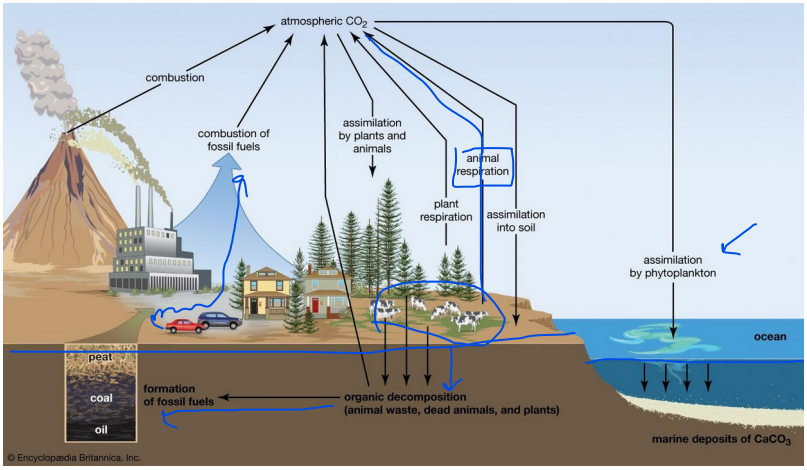


Fig: The Carbon Cycle.

NITROGEN CYCLE

Q6. Describe the processes involved in nitrogen cycle. [7]

Answer: The nitrogen cycle is the biogeochemical cycle by which nitrogen is converted into multiple chemical forms as it circulates among atmospheric, terrestrial and marine ecosystems. Although 78% of the atmosphere is N_2, this form has limited biological availability, leading to scarcity of usable nitrogen.

- **Nitrogen Fixation:** Conversion of inert atmospheric N_2 into ammonia/ammonium by nitrogen-fixing bacteria (e.g., Rhizobium in root nodules, free-living bacteria in soil), lightning, or industrial fertilizer production.
- **Ammonification:** Decomposers (bacteria and fungi) break down dead plants/animals and nitrogenous waste (urine), releasing ammonium (NH_4^+)/ammonia (NH_3) into the soil.
- **Nitrification:** Nitrifying bacteria oxidize ammonium to nitrite (NO_2^-) and then to nitrate (NO_3^-).
- **Assimilation:** Plants take up ammonium and nitrate from the soil and incorporate them into organic molecules such as proteins; animals assimilate nitrogen by consuming plants.
- **Denitrification:** Denitrifying bacteria convert nitrate back into atmospheric N_2, completing the cycle.

Human activities — fossil fuel combustion, use of artificial nitrogen fertilizers and release of nitrogen in wastewater — have dramatically altered the global nitrogen cycle, which can negatively affect both the natural environment and human health. Since nitrogen availability affects primary production and decomposition rates, the nitrogen cycle is of particular interest to ecologists.

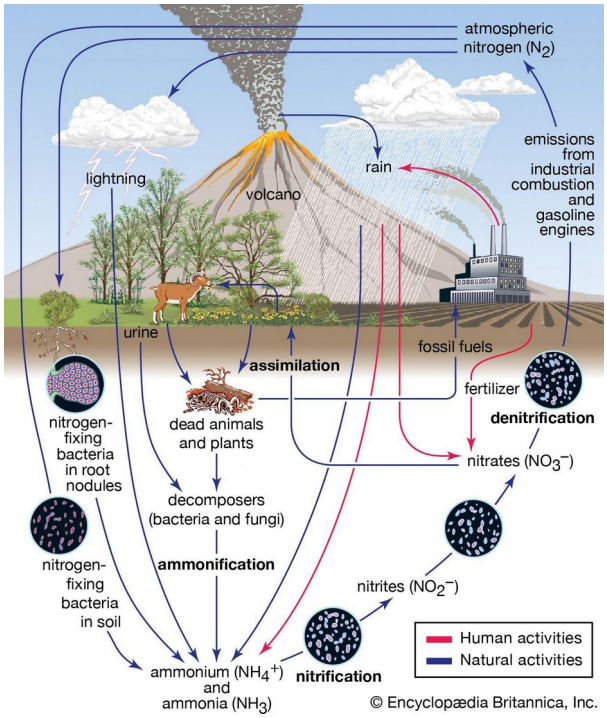


Fig: The Nitrogen Cycle — fixation, ammonification, nitrification, assimilation and denitrification.

ENVIRONMENTAL IMPACT ASSESSMENT

Q7. Discuss Environmental Impact Assessment. [7]

Answer: Environmental Impact Assessment (EIA) is a process that requires consideration of the environment and public participation in the decision-making process of project development. An Environmental Impact Statement (EIS) is a review document prepared for assessment in the EIA process.

The stages of EIA include the following:

- **1. Screening:** Decides which projects should be subject to environmental assessment. Criteria include threshold, size of the project, and sensitivity of the environment.
- **2. Scoping:** The process which defines the key issues that should be included in the environmental assessment.
- **3. EIS Preparation:** The scientific and objective analysis of the scale, significance and importance of the impacts identified.
- **4. Review:** The environmental assessment is reviewed by a government agency or an independent review panel, which guides the study and then advises decision-makers.

PHOSPHORUS AND SULFUR CYCLE

Q8. Discuss the Phosphorus and Sulfur cycle. [7]

Phosphorus Cycle

- Phosphorus is released from phosphate rocks through weathering and erosion, and by mining/extraction; it moves as runoff into rivers and streams.
- Phosphate is taken up by plants directly from soil or applied as fertilizer; it also dissolves in water bodies as dissolved phosphates, taken up by algae and other photosynthetic/aquatic organisms.
- Organic decomposition of animal waste and decaying plants/animals (on land and in water) returns phosphorus to the soil/water; in water bodies, unused phosphorus settles as marine sedimentation, eventually reforming phosphate rock over geological time.
- Unlike carbon or nitrogen, phosphorus has no significant atmospheric/gaseous phase — it is therefore called a sedimentary cycle, and weathering of rock is its principal natural source.

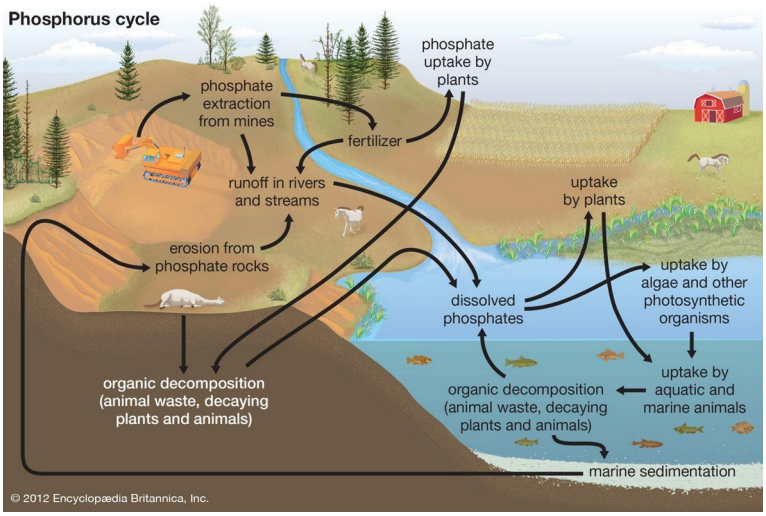


Fig: The Phosphorus Cycle.

Sulphur Cycle

- Sulphur enters the atmosphere as SO2 and H2S from natural sources (volcanic activity, decomposition of living things and waste products) and anthropogenic sources (smelting and burning of fossil fuels).
- Sulphur compounds return to Earth as wet deposition (acid rain, snow) or dry deposition, and as organic deposition; sulfates accumulate in soil ($SO4^{2-}$) and in water.
- Microorganisms in soil and water convert sulfur compounds between forms; sulfates and sulfides may be deposited in sediments, and iron sulfides accumulate in sediment, eventually forming ores/fossil fuels over geological time.

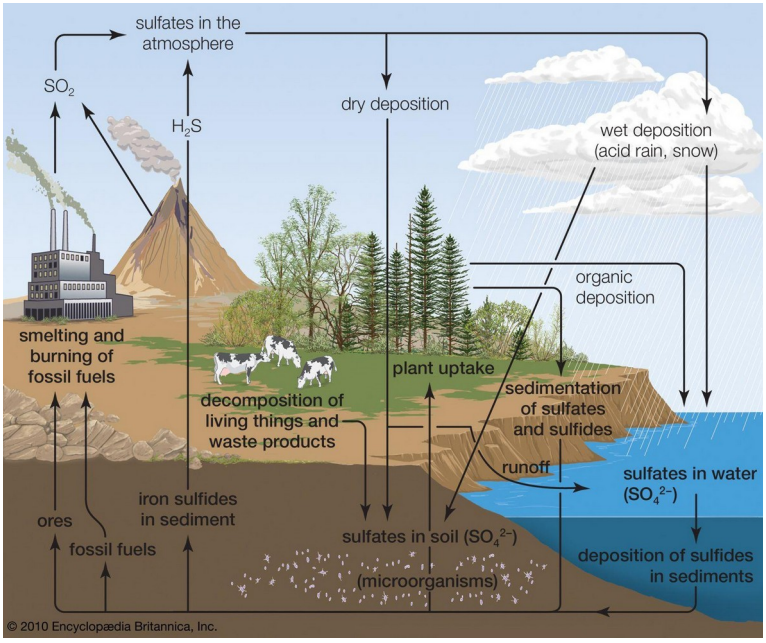


Fig: The Sulphur Cycle.

OXYGEN CYCLE

Q9. Describe the Oxygen cycle. [7]

Answer: The oxygen cycle describes the movement of oxygen through the atmosphere, biosphere, hydrosphere and lithosphere, closely linked with both the carbon and water cycles.

- Atmospheric O2 is released by photosynthesis in green plants (oxygen release from photosynthesis) and by rock weathering; CO2 is taken up by plants during photosynthesis.
- O2 is consumed through animal and plant respiration, and through fossil fuel/biomass combustion — both processes release CO2 back to the atmosphere.
- Water and oxygen exchange is also linked through evaporation, transpiration from plants and soil, condensation and precipitation — connecting the oxygen cycle with the hydrologic cycle.

- Decay of dead plants and animals also consumes oxygen (via decomposer respiration) and releases CO2.

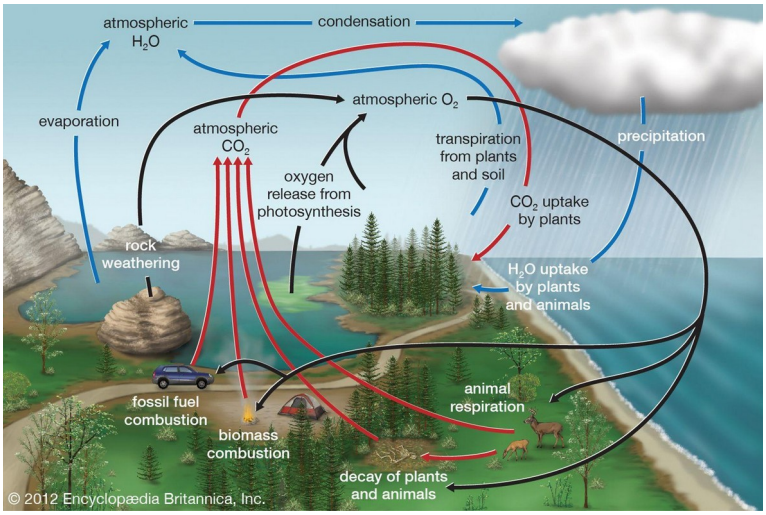


Fig: The Oxygen Cycle.

TOLERANCE CURVE

Q10. What do you mean by Tolerance Curve? Draw the Tolerance Curve and explain each part of the curve. [7]

Answer: Tolerance refers to the range of physical conditions (e.g., temperature, moisture, pH, altitude) within which an organism can survive. If we plot some measure of success of a species population (such as population density or fitness) against an environmental gradient, a tolerance curve is generated — also known as the "universal tolerance curve." It allows predictions about where a species will be found.

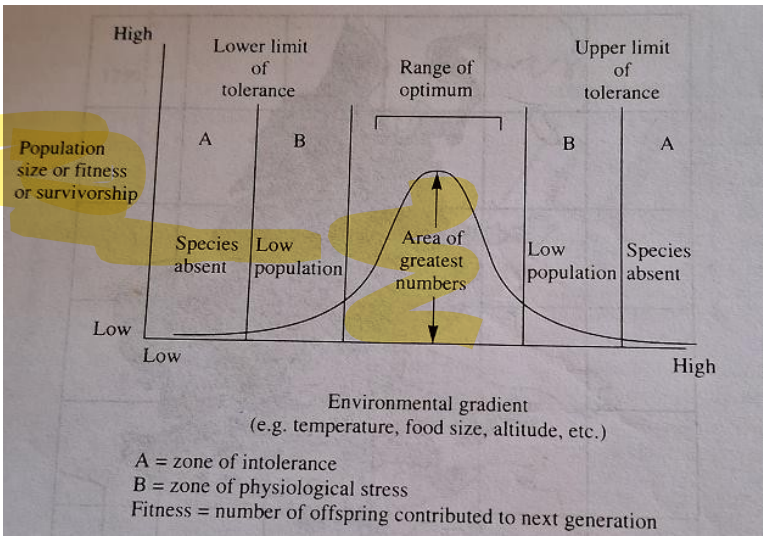


Fig: A schematic tolerance curve for a single species population existing on a single environmental gradient.

- **Range of Optimum:** Within this range, the majority of individuals of the species population can survive and reproduce; a large population size is maintained since conditions along this part of the gradient are ideal for the species (the "area of greatest numbers").
- **Zone of Physiological Stress:** Beyond the range of optimum, towards the high and low ends of the gradient, most individuals suffer increasing physiological stress. Conditions here are less than optimal — most individuals can stay alive but cannot function efficiently nor reproduce (marked as Zone B — low population).
- **Zone of Intolerance:** At the extremes of the gradient, no individuals of the species can survive because physical conditions or lack of food resources are too extreme to permit survival — the species is absent here (marked as Zone A).
- **Axes:** The x-axis represents the range of the environmental factor (low to high); the y-axis represents population size, fitness or survivorship (number of offspring contributed to the next generation).

HYDROLOGIC CYCLE

Q11. Describe the Hydrologic (Water) Cycle. [7]

Answer: The hydrologic cycle describes the continuous movement of water within the Earth and atmosphere, driven by solar energy. The ocean covers 71 percent of the Earth's surface (196,950,000 sq. mi / 510,000,000 sq. km) and is the largest reservoir of water.

- **Evaporation & Transpiration:** Water changes from liquid to vapour from oceans, water surfaces, land, soil and reservoirs; plants also release water vapour through transpiration from vegetation.
- **Condensation:** Water vapour rises and cools, forming clouds.
- **Precipitation:** Water falls back to Earth as rain/snow, either to land or directly to the ocean.
- **Runoff & Infiltration:** Water on land moves as surface runoff into rivers/streams, or infiltrates the soil (infiltration) to recharge the water table; water percolates deeper into the ground and moves as subsurface flow and groundwater outflow, eventually reaching the ocean (with saltwater intrusion possible near coastlines).

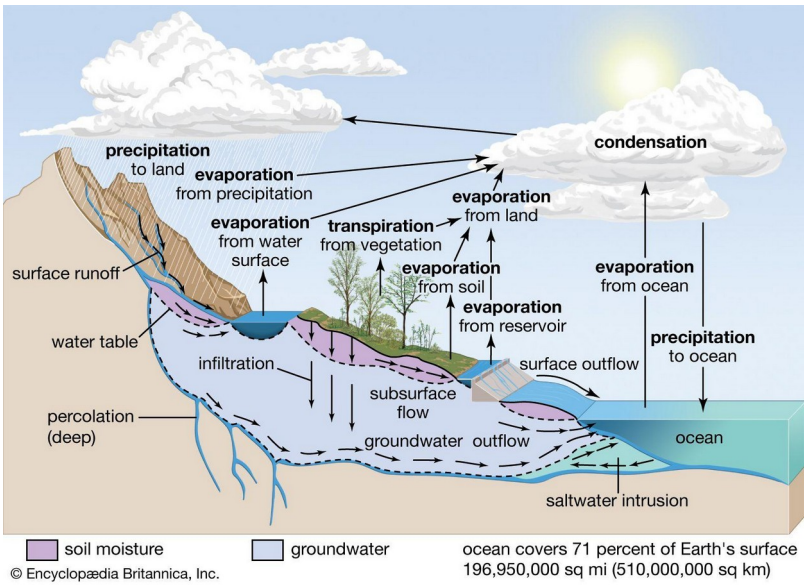


Fig: The Hydrologic (Water) Cycle.

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